



AI-assisted 30-day hospital readmission research dashboard

Hackathon prototype for research and demonstration only. Predictions are not clinically validated and must not guide real patient care.

1. Model Performance

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.6404	0.1654	0.5495	
Random Forest	0.6184	0.1623	0.5817	
XGBoost	0.8885	0.5357	0.0066	

Highest Recall

Random Forest

↑ 0.582

Highest Precision

XGBoost

↑ 0.536

Highest ROC_AUC

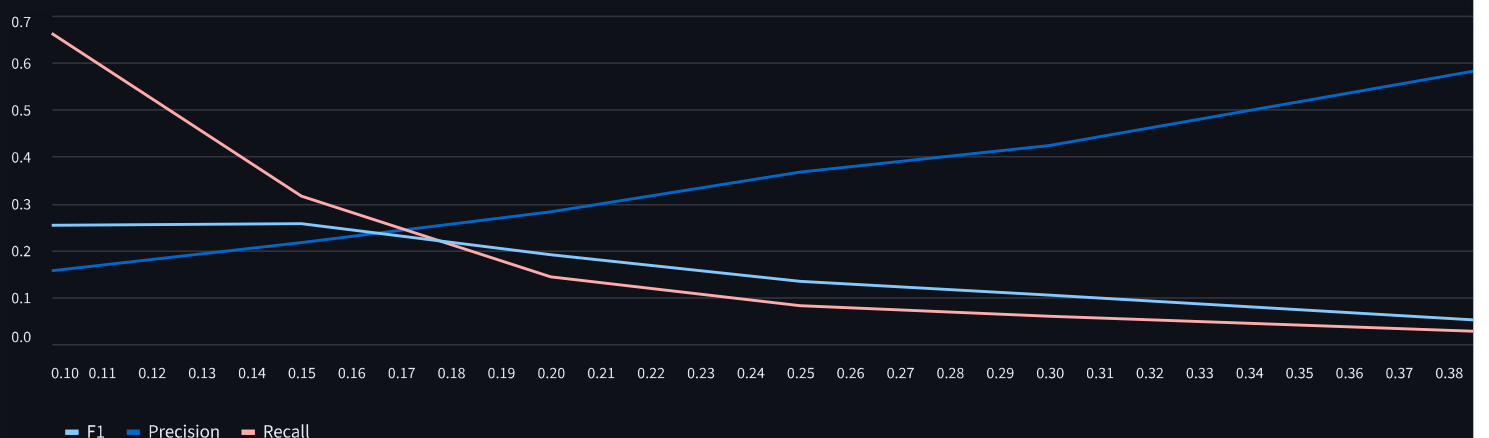
XGBoost

↑ 0.655

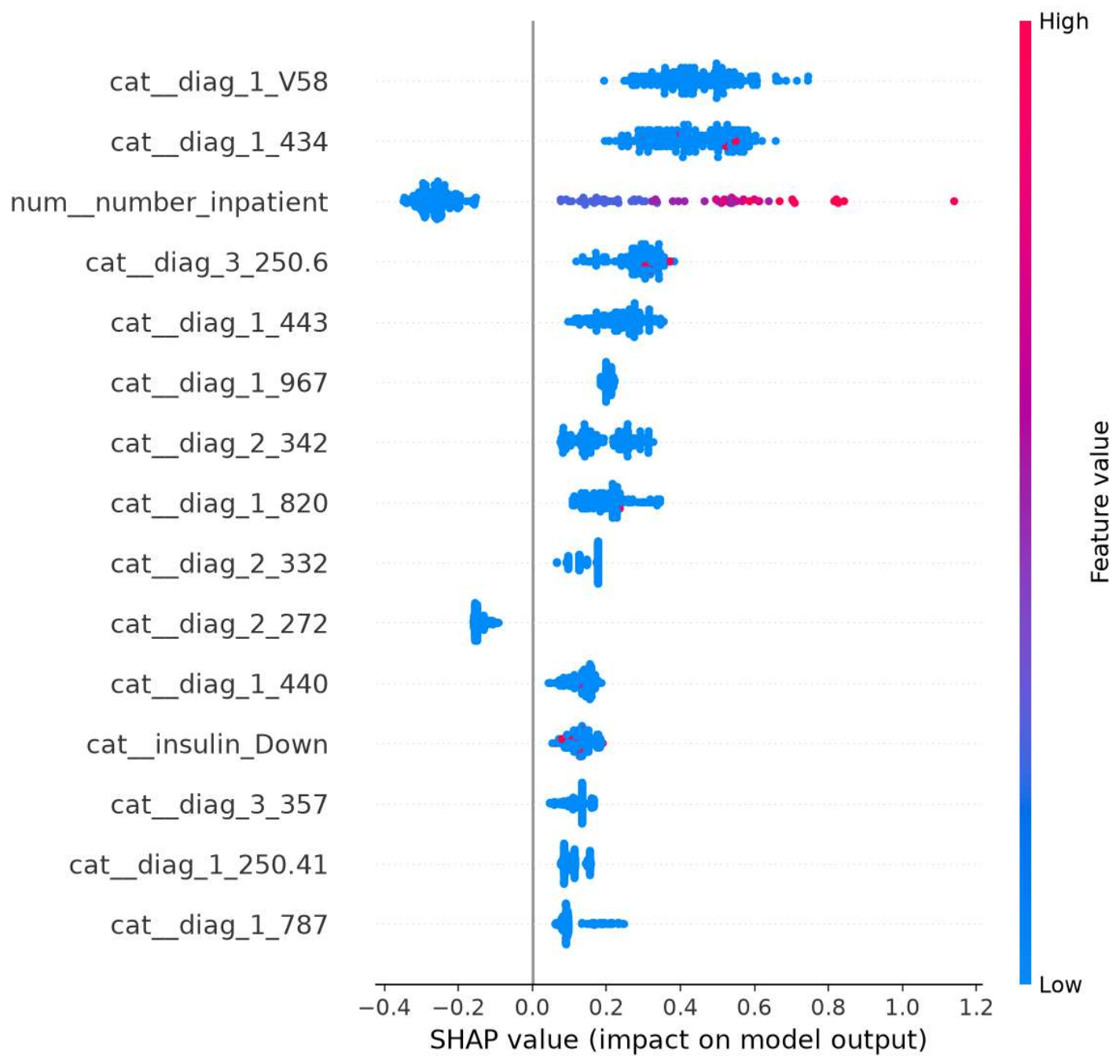
2. Readmission Threshold Analysis

Lower thresholds may detect more positive cases but can increase false positives. These are exploratory test-set results.

Threshold	Precision	Recall	F1
0.1	0.1564	0.6623	0.2581
0.15	0.2167	0.3157	0.2581
0.2	0.282	0.1435	0.1905
0.25	0.3669	0.0819	0.1333
0.3	0.4232	0.0594	0.1011
0.4	0.6098	0.022	0.0408
0.5	0.5357	0.0066	0.0099



3. Explainable AI



SHAP feature impact summary for a sample of test encounters

SHAP helps investigate which model features influence predictions. Feature influence does not establish causation.

4. Preliminary Fairness Audit

Group-level results depend on sample size and data quality. They do not prove the model is fair or unbiased.

Race	Sample_Size	Actual_Readmission_Rate	Predicted_High_Risk_Rate
AfricanAmerican	3866	0.1117	0.478
Asian	123	0.0813	0.349
Caucasian	15223	0.1128	0.47
Hispanic	404	0.1238	0.391
Other	276	0.0725	0.423
Unknown	462	0.0909	0.337

5. ReCare AI Workflow

1. Load the UCI diabetes hospital encounters dataset.
2. Prepare features and define the 30-day readmission outcome.
3. Train and compare Logistic Regression, Random Forest and XGBoost.
4. Explore threshold trade-offs between precision and recall.
5. Inspect model explanations with SHAP.
6. Review performance across demographic groups.

Prototype status: educational research demo — not for diagnosis, treatment decisions, or deployment in clinical workflows.